

The Biggest Mistake Hotels Are Making About AI

The industry doesn't have an adoption problem. It has an architecture problem.

By Dr. Tong Yin

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The hotel industry does not have an AI adoption problem.

It has an AI misunderstanding problem.

Across global markets, hotels are deploying AI at unprecedented speed — chatbots, pricing engines, voice assistants, analytics dashboards. On the surface, the industry appears transformed.

Underneath, very little has changed.

AI Theatre

I call this phenomenon AI Theatre: the deployment of visible technology that creates the impression of transformation without changing the underlying economics of the business.

A chatbot that answers FAQs but cannot complete a booking, resolve a complaint, or escalate intelligently. Its primary function is to demonstrate to the owner that the hotel “has AI.”

Booking conversion does not change. A pricing system whose recommendations the revenue manager silently overrides more than half the time — without the system learning from those overrides. Revenue does not measurably improve. The owner is told “AI is now driving revenue.”

A sentiment dashboard that aggregates reviews from every platform into a single display that nobody reads daily and that produces no changed behaviour.

In each case, AI has been deployed at the level of feature, not at the level of operational architecture. The features are visible to owners, to boards, to investors. The architecture — where genuine transformation happens — remains untouched.

This pattern is not confined to any single market. I have observed it across Asia-Pacific, the Middle East, and increasingly in European and North American operations. The technology varies. The architectural gap is universal.

The Override Problem

The clearest signal of architectural mismatch is override behaviour.

Across multiple markets I have studied, experienced revenue managers routinely override AI pricing recommendations. Often more than half the time. The override is silent. The system does not learn. The human does not trust. And the hotel pays a five-figure annual SaaS bill for a system whose recommendations are mostly being ignored.

This is not a usability problem. It is a design failure.

When an experienced operator overrides a recommendation, they are expressing tacit knowledge the system does not yet possess — about a local event the data has not captured, a shift in source-market sentiment, or a competitive dynamic that defies historical pattern. If that knowledge is captured, it compounds into competitive advantage over twelve to eighteen months. If it is not captured, it evaporates when the revenue manager changes jobs.

Most systems are designed to gradually grind down override behaviour until the human stops bothering. That is precisely backward. The system should be designed to absorb the override as its most valuable input.

What Architecture Actually Means

True transformation requires three structural changes, not three new features.

First, automating work that should have been automated a decade ago — rate parity monitoring, inventory synchronisation, housekeeping route optimisation, predictive maintenance triage. This is unglamorous. It is also where 8–15 per cent of operational margin typically sits.

Second, augmenting human judgement rather than replacing it. The General Manager, Director of Sales, Executive Chef — these roles exist because human judgement is the product. AI should expand the space in which the human applies judgement, not compress it.

Third, designing systems that learn from human input over time. Not static automation. Adaptive intelligence.

The Real Divide

The divide in the next decade will not be between hotels that use AI and those that do not. Every hotel will use AI. That distinction is already meaningless.

The divide will be between those that redesign operations around AI — automating what should be automated, augmenting what should remain human, and building systems that learn from the humans who use them — and those that decorate existing systems with AI features that look impressive in board presentations and produce no measurable change.

The hotels that get this right will not be the most automated hotels in the world. They will be the most intelligently human. That is a distinction the market will reward.

AI does not create competitive advantage. Architecture does.

About the author

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